

Hand-in assignment 4

Instructions

- This assignment consists of **two questions**.
- On Vula it opens at 12:00 on Friday, 10 May 2024, please upload your solutions under **Vula Assignments no later than 23:55 on Tuesday, 14 May 2024**. Make sure that no pages or questions are missing.
- Write your full name and student number on your solutions.
- If possible, **upload a single document** saved as **ABCXYZ001_A4** (where **ABCXYZ001** is your campus Id).
- Show all your work or reasoning. **Answers without work or reasoning shown where needed will receive no mark.**

Questions

1. State whether the following are true or false. If true, prove it or if false, give a counter-example.
 - a) Let $(A_i)_{i \in I}$ be a family of compact subsets of metric space (X, d) , then $\cup_{i \in I} A_i$ is compact. That is, arbitrary union of compact subsets is compact. [3]
 - b) Let $(A_i)_{i \in I}$ be a family of compact subsets of metric space (X, d) , then $\cap_{i \in I} A_i$ is compact. That is, arbitrary intersection of compact subsets is compact. [3]
 - c) Let (X, d_X) and (Y, d_Y) be metric spaces, $f : X \rightarrow Y$ a continuous function. If $B \subseteq Y$ is compact, then the preimage $f^{-1}(B)$ is compact. [3]
 - d) Let $(A_i)_{i \in I}$ be a family of connected subsets of metric space (X, d) , then $\cup_{i \in I} A_i$ is connected. That is, arbitrary union of connected subsets is connected. [3]
 - e) Let $(A_i)_{i \in I}$ be a family of connected subsets of metric space (X, d) , then $\cap_{i \in I} A_i$ is connected. That is, arbitrary intersection of connected subsets is connected. [3]
2. Prove that a metric space (X, d) is compact if and only if for any decreasing sequence (A_n) of non-empty subsets of X , $\cap_{n \in \mathbb{N}} \overline{A_n} \neq \emptyset$. [6]